

Fast Thermal Modeling of TTSV via Bounded Neural Networks

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Abstract—Effective thermal management in chiplet-based architectures necessitates accurate prediction of Thermal Through-Silicon Via (TTSV) anisotropic equivalent thermal conductivity. This paper introduces a novel physics-informed hybrid model to efficiently determine both vertical and particularly the challenging horizontal thermal conductivity components. The approach integrates an analytical Series-Parallel Thermal Conductivity model, which calculates vertical conductivity and defines rigorous physical bounds for horizontal conductivity, with a Back-Propagation Neural Network. This network, featuring a bounded output layer constrained by these physical limits, accurately predicts horizontal thermal conductivity, achieving a mean absolute percentage error of 0.35 percent or less compared to finite element method benchmarks. The proposed model demonstrates significantly improved accuracy over traditional methods and faster training convergence, offering a robust tool for advanced packaging thermal simulations.

Keywords—Thermal through-silicon via, Equivalent thermal conductivity, Neural Network, Thermal Management

I. INTRODUCTION

Chiplet-based architectures are key to enhancing microelectronic system performance and functionality, overcoming monolithic integration limitations [1-3]. Assembling multiple specialized chiplets on a common platform provides these systems with significant advantages in design flexibility, yield, and heterogeneous integration. However, this dense integration concurrently exacerbates thermal management challenges, with high power densities and complex heat dissipation pathways potentially leading to performance degradation and reliability concerns [4-6]. Thermal Through-Silicon Vias (TTSVs) have become indispensable components in advanced packaging, creating crucial vertical pathways for efficient heat removal from active chiplets [7-9].

A critical factor underpinning the effective thermal design and simulation of chiplet systems incorporating TTSVs is the accurate determination of their anisotropic equivalent thermal conductivity (k_{eq}). Although the vertical k_{eq} can often be reasonably estimated using established thermal resistance models [10-11], the prediction of horizontal k_{eq} presents a more formidable challenge. Horizontal heat flow is characterized by intricate interactions among the highly conductive via material, the dielectric liner, and the surrounding substrate. Consequently, the horizontal k_{eq} is frequently neglected [12-13] or significantly oversimplified using rudimentary approximations [14-16]. These inaccuracies can propagate through thermal simulations, leading to unreliable temperature predictions, erroneous hotspot identification, and consequently,

suboptimal thermal management strategies. Conventional finite element method (FEM) simulations are capable of accurately capturing these effects; however, they are computationally intensive, which hinders their application in rapid parametric studies or integration into iterative design optimization loops.

To address the critical need for an accurate and efficient method to determine the anisotropic k_{eq} of TTSVs, particularly their horizontal component, this paper proposes a novel physics-informed hybrid modeling approach. The methodology underpinning the present study combines the strengths of an analytical Series-Parallel Thermal Conductivity (SPTC) model with a Back-Propagation Neural Network (BPNB). The SPTC model is used to directly calculate the vertical k_{eq} and, importantly, to establish rigorous physical upper and lower bounds for the horizontal k_{eq} . Informed by these bounds, the BPNB model, featuring a tailored architecture with a parametric sigmoid activation function in the hidden layer and a physically constrained sigmoid activation function in the output layer, is trained to precisely predict the horizontal k_{eq} within these physically meaningful limits.

This paper is organized as follows: Section II details the formulation of the SPTC model and the derivation of physical bounds for horizontal thermal conductivity. In addition, this section describes the architecture, training, and innovative bounded-output mechanism of the BPNB model. Section III presents the experimental validation of the proposed hybrid approach, demonstrating its accuracy. Section IV concludes the paper.

II. HYBRID MODELING APPROACH FOR TTSV EQUIVALENT THERMAL CONDUCTIVITY PREDICTION

To accurately and efficiently determine the anisotropic k_{eq} of TTSVs, this study employs a hybrid modeling approach. The present methodology is predicated on a synergistic combination of the SPTC model and the BPNB model. The SPTC model is employed to directly compute the vertical equivalent thermal conductivity (k_z) and to establish rigorous, physical bounds for the horizontal equivalent thermal conductivity (k_x). Subsequently, the BPNB model, informed by these bounds, is employed to precisely predict k_x .

A. Series-Parallel Equivalent Thermal Conductivity Model

The SPTC model analyzes a representative cylindrical TTSV unit, as depicted schematically in Fig. 1. This composite structure typically comprises a central core of highly conductive material (Cu), enveloped by a thin dielectric insulating layer (SiO_2), all embedded within a substrate material (Si). The key geometrical parameters influencing its k_{eq} include the metal core radius, the dielectric layer thickness, the overall height of the TTSV, and the pitch or side length of the square unit cell. For analytical tractability, the TTSV unit is conceptually divided along its

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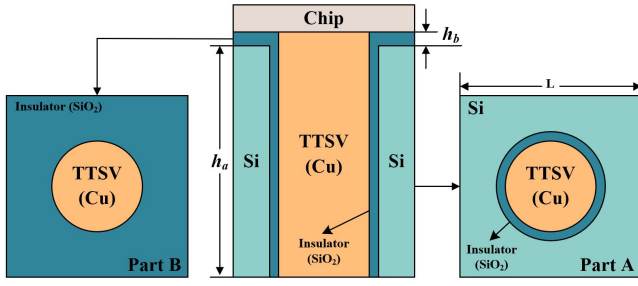


Fig. 1. Schematic diagram of cylindrical TTSV.

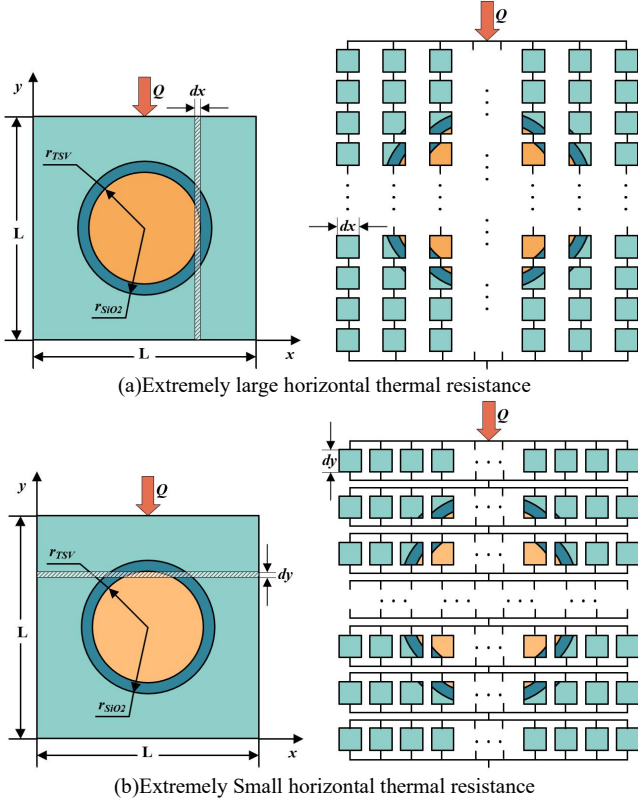


Fig. 2. Thermal resistance network model for Part A of TTSV unit

height into two distinct segments: Part A represents the main TTSV structure with height h_a , and Part B is typically a thinner section, an oxide isolation layer with height h_b . The thermal conductivities of these parts are analyzed individually before being combined.

The vertical k_{eq} (k_z) of the TTSV unit is computed using a parallel thermal resistance model, where the contribution of each material is weighted by its respective volume fraction. For Part A of the TTSV unit, its vertical equivalent thermal conductivity k_{z_A} can be calculated as follows:

$$\begin{cases} k_{z_A} = k_{Cu} v_{Cu_A} + k_{SiO_2} v_{SiO_2_A} + k_{Si} v_{Si_A} \\ v_{Cu_A} = \frac{\pi r_{TTSV}^2}{L^2}, v_{SiO_2_A} = \frac{\pi r_{SiO_2}^2 - \pi r_{TTSV}^2}{L^2} \\ v_{Si_A} = 1 - v_{TTSV_A} - v_{SiO_2_A} \end{cases} \quad (1)$$

where k_{Cu} , k_{SiO_2} and k_{Si} are the thermal conductivities of Cu, SiO₂ and Si respectively, r_{SiO_2} and r_{TTSV} are the radii of TTSV oxide and TTSV metal respectively, v_{Cu_A} , $v_{SiO_2_A}$ and v_{Si_A} are the volume fractions of Cu, SiO₂ and Si in Part A respectively, and L is the side length of the square TTSV unit. Similarly, the vertical equivalent thermal conductivity k_{z_B} of

part B is determined using a similar method based on the volume fraction of its constituent materials. The overall vertical equivalent thermal conductivity k_z of the entire TTSV unit is then determined by treating Parts A and B as thermal resistances in series:

$$\frac{1}{k_z} = \frac{h_a}{k_{z_A}} + \frac{h_b}{k_{z_B}} \quad (5)$$

The determination of horizontal equivalent thermal conductivities k_x and k_y is more intricate due to the two-dimensional heat flow paths that involve both series and parallel resistances. Given the cylindrical symmetry of the TTSV within a square unit cell, the horizontal thermal conductivity is isotropic $k_x = k_y$. The SPTC model provides a robust estimation of the range for k_x by establishing theoretical lower and upper bounds. This is achieved by considering extreme cases of series-parallel thermal resistance arrangements for heat flow in the lateral direction within Part A, as illustrated by the equivalent network models in Fig. 2.

1) The series-parallel (s-p) model corresponds to the maximum lateral thermal resistance case. It is assumed that the heat flow passes preferentially in series through the interfaces of the different materials and then in parallel as a whole. The resulting lower limit of the horizontal equivalent thermal conductivity k_{A_s-p} , is calculated as follows:

$$\begin{cases} k_{A_s-p} = 2 \int_0^{r_{TTSV}} \frac{dx}{\frac{2l_{TTSV}}{k_{Cu}} + \frac{2l_{SiO_2} - 2l_{TTSV}}{k_{SiO_2}} + \frac{L - 2l_{SiO_2}}{k_{Si}}} \\ + 2 \int_{r_{TTSV}}^{r_{SiO_2}} \frac{dx}{\frac{2l_{SiO_2}}{k_{SiO_2}} + \frac{L - 2l_{SiO_2}}{k_{Si}}} + \frac{k_{Si}(L - 2r_{SiO_2})}{L} \\ l_{TTSV} = \sqrt{r_{TTSV}^2 - x^2}, l_{SiO_2} = \sqrt{r_{SiO_2}^2 - x^2} \end{cases} \quad (2)$$

2) The parallel-series (p-s) model corresponds to the minimum lateral thermal resistance case. It is assumed that the heat flow passes preferentially in parallel through each material layer and then in series as a whole. This results in an upper limit of the horizontal equivalent thermal conductivity k_{A_p-s} , calculated as follows:

$$\begin{cases} \frac{1}{k_{A_p-s}} = 2 \int_0^{r_{TTSV}} \frac{dy}{2k_{Cu} l_{TTSV} + 2k_{SiO_2}(l_{SiO_2} - l_{TTSV}) + k_{Si}(L - 2l_{SiO_2})} \\ + 2 \int_{r_{TTSV}}^{r_{SiO_2}} \frac{dy}{2k_{SiO_2} l_{SiO_2} + k_{Si}(L - 2l_{SiO_2})} + \frac{L - 2r_{SiO_2}}{L k_{Si}} \\ l_{TTSV} = \sqrt{r_{TTSV}^2 - y^2}, l_{SiO_2} = \sqrt{r_{SiO_2}^2 - y^2} \end{cases} \quad (3)$$

Therefore, the actual horizontal equivalent thermal conductivity of Part A k_{x_A} must lie within the interval $[k_{A_s-p}, k_{A_p-s}]$. Similarly, applying these assumptions to Part B yields bounds for its horizontal equivalent thermal conductivity k_{x_B} . It must lie within the interval $[k_{B_s-p}, k_{B_p-s}]$. Assuming Parts A and B are connected in parallel horizontally, the overall bounds $[k_{s-p}, k_{p-s}]$ for the horizontal equivalent thermal conductivity k_x of the entire TTSV unit are derived

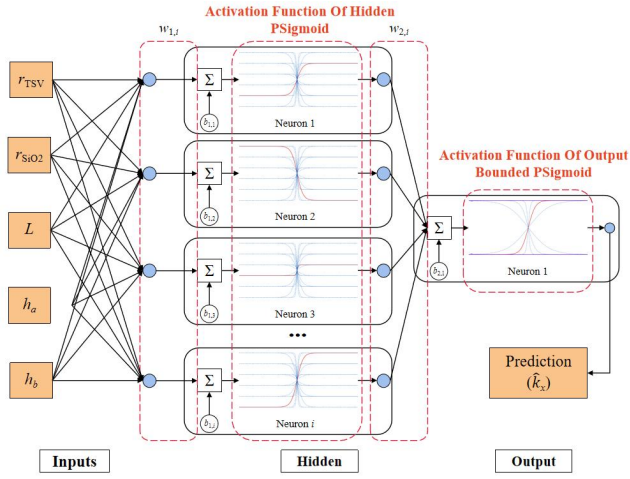


Fig. 3. Architecture of the BPNB model with bounded output

by combining the respective bounds of Parts A and B:

$$\begin{cases} k_{s-p} = \frac{h_a}{h_a + h_b} k_{A-s-p} + \frac{h_b}{h_a + h_b} k_{B-s-p} \\ k_{p-s} = \frac{h_a}{h_a + h_b} k_{A-p-s} + \frac{h_b}{h_a + h_b} k_{B-p-s} \end{cases} \quad (4)$$

Thus, the SPTC model provides a direct calculation for k_z and establishes the strict physical bounds for k_x . These bounds are crucial as they inform and constrain the subsequent BPNB model, ensuring physically plausible predictions and enhancing the learning efficiency of the neural network.

B. Back-Propagation Neural Network Model with Bounded Output

The architecture of the BPNB model is illustrated in Fig. 3. The structure of the network is as follows: the input layer contains five nodes corresponding to the key structural parameters of the TTSV cells that significantly affect k_x . The number of nodes in the hidden layer is based on Kolmogorov's theorem, which suggests approximately $2n+1$ hidden nodes for n inputs. The output layer comprises a single node that predicts the horizontal equivalent thermal conductivity k_x . Training data for the BPNB model, comprising sets of TTSV structural parameters and their corresponding k_x values, were generated using FEM simulations. The mathematical formulation of the BPNB model can be expressed as:

$$\hat{k}_x = g(w_2 \cdot f(w_1 x + b_1) + b_2) \quad (5)$$

where x is the input vector of TTSV structural parameters, w_1 and w_2 are the weight matrices for the hidden and output layers, respectively, and b_1 and b_2 are the corresponding bias terms. $f(x)$ represents the activation function of the hidden layer, and $g(x)$ is the activation function of the output layer.

A pivotal innovation of the BPNB model is the selection and parametrization of its activation functions. The hidden layer activation function employs a parametric Sigmoid (PSigmoid) function, which is defined as follows:

$$f(y_i) = \frac{d_i}{1 + e^{-a_i(y_i + b_i)}} + c_i \quad (6)$$

where y_i is the input of the activation f on the i th channel, with learnable parameters a_i , b_i , c_i , and d_i controlling the shape and properties of the sigmoid curve. This allows the PSigmoid function to adapt its shape dynamically during training, offering significantly greater non-linear expressive capacity compared to a standard sigmoid function where these parameters are fixed.

In regression problems, a linear activation function is commonly used for the output layer. Combine or rephrase: However, the horizontal equivalent thermal conductivity k_x of the TTSV unit must be physically regular and fall within the specific range derived from the SPTC model. In this study, the PSigmoid function is innovatively adopted as the activation function of the output layer, with the physical boundary being directly embedded in it. Its specific form is given by:

$$\hat{k}_x = g(y) = \frac{k_{p-s} - k_{s-p}}{1 + e^{-a(y+b)}} + k_{s-p} \quad (7)$$

The physical constraints are embedded by setting the PSigmoid parameters c and d as fixed constants derived from the SPTC model: the lower bound $c = k_{s-p}$ and the range width $d = k_{p-s} - k_{s-p}$. These fixed values mathematically constrain the output within the physical boundaries $[k_{s-p}, k_{p-s}]$. Concurrently, parameters a and b remain learnable, allowing the network to adjust the Sigmoid shape during training to accurately model the input k_x relationship within these bounds.

Overall, the BPNB model, by operating within the physical boundaries established by the SPTC model and leveraging adaptive activation functions, provides an accurate and robust method for predicting the horizontal equivalent thermal conductivity k_x . When combined with the directly calculated vertical component k_z from the SPTC model, this hybrid approach yields the comprehensive anisotropic thermal properties essential for detailed and reliable thermal modeling of TTSV structures in advanced electronic packaging.

III. EXPERIMENTAL VALIDATION AND RESULT ANALYSIS

To rigorously evaluate the proposed hybrid modeling approach, this section focuses on validating the BPNB model's performance in predicting the horizontal equivalent thermal conductivity of TTSVs. Benchmark data, comprising TTSV structural parameters and corresponding FEM-simulated k_x values, were generated by systematically varying the metal core radius, SiO₂ dielectric layer thickness, and unit cell pitch. The predictive capabilities of the BPNB model were compared against the SPTC model's theoretical lower (k_{s-p}) and upper (k_{p-s}) bounds, their arithmetic average (k_{avg}) [17], a calibrated analytical model from literature [18], and a traditional back-propagation neural network (BPNN) with standard sigmoid and linear activation functions.

The results demonstrate the superior accuracy of the BPNB model. As summarized in Table I, the BPNB model achieved a remarkably low MAPE, typically between 0.310% and 0.350% across various parameter sweeps, significantly outperforming the standalone SPTC bounds and the k_{avg} approximation. For instance, when varying the metal radius, the BPNB model yielded a MAPE of 0.310% and a maximum percentage error of 0.357%, whereas k_{avg} resulted

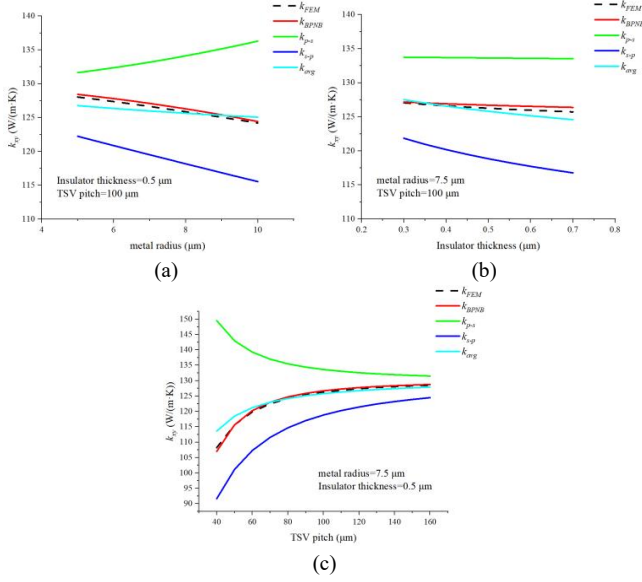


Fig. 4. Comparison of BPNB predicted k_x values against FEM simulations for varying (a) metal radius, (b) insulator thickness, and (c) TTSV pitch.

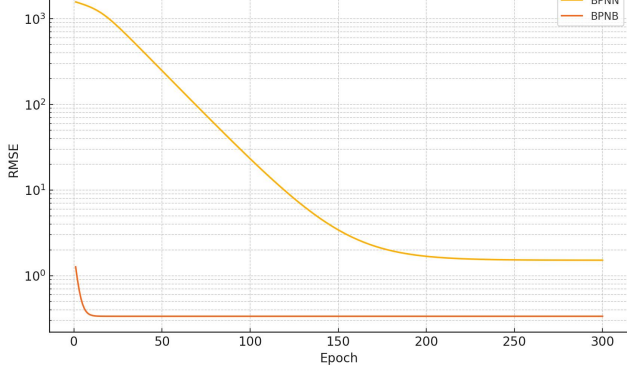


Fig. 5. Comparison of convergence speed between the proposed BPNB model and a traditional BPNN.

in a MAPE of 0.530% and the analytical model [18] reported errors around 3%. Fig. 4 further illustrates the excellent agreement between BPNB predicted k_x values and FEM benchmarks across variations in key geometrical parameters, highlighting its ability to capture complex non-linear dependencies.

To evaluate the advantages of the proposed BPNB model, its training process and final accuracy were compared with those of a conventional BPNN. Both networks share the same architecture and were trained on the same FEM generated dataset using identical optimization algorithms and hyperparameters. Unlike the BPNB model, the conventional BPNN employs a standard sigmoid activation function in the hidden layer and a linear activation function in the output layer, and thus includes no physically based boundary constraints.

Fig. 5 plots the RMSE loss versus the number of training epochs for both networks. The BPNB model demonstrates a faster convergence rate and achieves a substantially lower final RMSE, reached within 1% of the final converged RMSE loss value of 0.34 in a mere tens of epochs. Conversely, the conventional BPNN requires hundreds of epochs to reach a comparatively high RMSE value of 1.52. Consequently, the final converged RMSE of the BPNB model is more than four times lower than that of the traditional BPNN.

TABLE I. COMPARISON OF TTSV k_x PREDICTION METHODS

Method	Parameter Varied	Performance metrics			
		MAPE (%)	RMSE	Max Error	Max Error (%)
Proposed (k_{BPNB})	Metal radius	0.310	0.397	0.453	0.357
	SiO ₂ thickness	0.334	0.457	0.662	0.527
	Pitch	0.350	0.501	1.224	1.131
Series-Parallel Bound ($k_{\text{s-p}}$)	Metal radius	5.825	7.397	8.660	6.972
	SiO ₂ thickness	5.767	7.380	8.956	7.124
	Pitch	7.067	9.382	16.593	15.328
Parallel-Series Bound ($k_{\text{p-s}}$)	Metal radius	6.020	8.036	12.090	9.733
	SiO ₂ thickness	5.803	7.338	7.814	6.215
	Pitch	10.183	16.225	41.281	38.133
Bound average [17] (k_{avg})	Metal radius	0.530	0.772	1.270	0.992
	SiO ₂ thickness	0.436	0.653	1.133	0.901
	Pitch	0.960	1.794	5.401	4.989
Calibrated analytical [18]	Overall	N/A	N/A	N/A	3

In summary, the experimental validation confirms that the proposed physics-informed BPNB model provides a highly accurate and efficient method for predicting the horizontal equivalent thermal conductivity of TTSVs. Its ability to integrate physical bounds with a data-driven approach results in superior performance compared to purely analytical models, empirical approximations, and traditional neural network architectures, making it a valuable tool for reliable thermal simulations in advanced packaging.

IV. CONCLUSION

This paper presented a novel physics-informed hybrid model to accurately and efficiently predict the anisotropic equivalent thermal conductivity (k_{eq}) of TTSVs, with a particular focus on its challenging horizontal component (k_x). The proposed approach integrates an analytical SPTC model with BPNB in a synergistic manner. The SPTC model directly calculates the vertical k_{eq} (k_z) and establishes rigorous physical bounds for k_x . The BPNB, featuring parametric sigmoid activation functions and an output layer explicitly constrained by these SPTC derived physical limits, demonstrated superior accuracy in predicting k_x , achieving a MAPE of 0.35% or less against FEM benchmarks. Furthermore, the BPNB model exhibited significantly faster convergence and lower training error compared to traditional neural network architectures. This work provides a robust and computationally efficient methodology for enhancing the fidelity of thermal simulations in advanced packaging, contributing to more reliable thermal management strategies for chiplet-based systems.

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